

Test Suite Refactoring and Integration Test Expansion

Overview

Refactored the integration test project to improve maintainability, scalability, and alignment with the formal test plan. The previous single `CustomerControllerTests.cs` file has been reorganized into endpoint-specific test classes and shared helper utilities.

Project Structure

```
DemoERPApi.Tests/  
|  
├─ Helpers/  
│  ├── JwtTestTokenHelper.cs  
│  ├── TestAuthHelper.cs  
│  └── TestData.cs  
|  
├─ Integration/  
│  ├── AuthControllerTests.cs  
│  ├── CustomerSyncTests.cs  
│  ├── CustomerGetTests.cs  
│  ├── CustomerUpdateTests.cs  
│  └── CustomerDeleteTests.cs
```

Changes

Test Organization

- Split integration tests into endpoint-specific files:
 - `AuthControllerTests.cs`
 - `CustomerSyncTests.cs`
 - `CustomerGetTests.cs`
 - `CustomerUpdateTests.cs`
 - `CustomerDeleteTests.cs`
- Added shared helper classes for authentication and reusable test data:
 - `JwtTestTokenHelper.cs`
 - `TestAuthHelper.cs`
 - `TestData.cs`

Authentication Improvements

- Enhanced JWT generation to support configurable usernames and roles.

- Added reusable methods for configuring test authentication scenarios:
 - Admin user
 - Authorized owner user
 - Unauthorized user
 - Invalid token
 - Missing token

Test Naming and Assertions

- Updated test names to accurately reflect expected outcomes.
- Removed ambiguous assertions and comments.
- Adopted self-documenting test names following the pattern:

GetCustomer_ReturnsNotFound_WhenAdminRequestsNonExistingCustomer()

Test Coverage Expansion

Implemented a comprehensive integration test suite consisting of 44 test cases:

Endpoint	Test Cases
Authentication (POST /api/Auth/login)	5
Customer Sync (POST /api/Customer/sync)	13
Customer Get (GET /api/Customer/{crmId})	8
Customer Update (PUT /api/Customer/update)	12
Customer Delete (DELETE /api/Customer/{crmId})	6
Total	44

Benefits

- Improved test maintainability and readability.
- Clear separation of responsibilities by endpoint.
- Reduced duplication through shared helpers and test data.
- Consistent handling of authorization scenarios.
- Better alignment between implementation and the formal test plan.
- Easier future expansion of integration test coverage.